



Machine Learning Task – 1

IEEE SSCS – Level 1

Task 1 (Easy)

Derive the gradient of the Binary Cross-Entropy (BCE) loss function with respect to each model weight, i.e., compute $\frac{\partial J}{\partial w_j}$

where J is the Binary Cross-Entropy cost function. Your derivation should be mathematically complete, starting from the BCE equation and applying all required differentiation steps until the final gradient expression is obtained. Clearly show every intermediate step, including the use of the chain rule where necessary. This task must be completed by hand on paper and submitted as a handwritten solution.

Task 2 (Easy)

Using NumPy, implement the Sigmoid function and generate its plot over a suitable range of input values. Your implementation should clearly demonstrate the behavior of the Sigmoid curve. Submit the Python source code along with the generated plot. Ensure your code is well-organized, properly commented, and produces a correctly labeled figure.

Task 3 (Medium)

Implement a reusable Logistic Regression model from scratch using Python functions only (without machine learning libraries). Your solution must follow a modular software design by using two separate Python files. The first file should contain all functions required for the Logistic Regression algorithm, including initialization, training, prediction, and evaluation. The second file should serve as the main program, where the

functions are imported and used to train and test the model. The implementation should be general enough to train Logistic Regression on any suitable binary classification dataset. Ensure your code is clean, modular, and well-documented.

You must follow this structure :

```
- main.py
- classification.py
  #contain LogisticRegression Class
```

```
class LogisticRegression :]
    def __init__(self, iterations, lr):
        self.iterations = iterations
        self.lr = lr
    def fit(x_train, y_train ) :
        # implement it

    def predict(x_test) :
        # implement it
```

Task 4 : Now test your model on simple data set (Medium)

Your dataset is 4D XOR truth table

<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>Out</i>
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0